PRODUCT USER MANUAL

For Global Ocean GMPE Sea Surface Temperature Multi Product Ensemble

SST_GLO_PHY_L4_NRT_010_005
Issue: 1.0

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RECORD TABLE

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GLOSSARY AND ABBREVIATIONS

CF	Climate Forecast (convention for NetCDF)
CHL	Chlorophyll
CMEMS	Copernicus Marine Environment Monitoring Service
GDS	GHRSST Data Specification
GHRSST	Group for High Resolution SST
GMPE	GHRSST Multi-Product Ensemble
MDS	Marine Data Store
MFC	Monitoring and Forecasting Centre
MFWAM	Meteo France WAve Model
MY	Multi Year
NetCDF	Network Common Data Form
NRT	Near Real Time
PUM	Product User Manual
QUID	Quality Information Document
TAC	Thematic Assembly Center
SIMULATION	An hindcast done without data assimilation.
NRT-ANALYSIS:	An hindcast done with data assimilation for a period very close to present. The quality and the number of available observations it's not optimal (less observations and/or less accurate than in a best-analysis).
BEST-ANALYSIS:	An hindcast done with data assimilation for a period close to present. The quality and the number of available observations it's the best available (more observations and/or more accurate than in a NRT-analysis).

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the Product User Manual (PUM) for the SST_GLO_PHY_L4_NRT_010_005 CMEMS (Copernicus Marine Environment Monitoring Service) product. It describes the process for building the product, content of the product, and data services from which the product is available to users. The product described here is also known to users as the GHRSST (Group for High Resolution SST) Multi-Product Ensemble (GMPE). It is produced over the global ocean and is processed at the Met Office.

The format of SST files within CMEMS follows that defined by the GHRSST project. GHRSST defined a new format 'GDS2.0', which the GMPE product transitioned to use in 2012. Hence the GMPE product was available under 2 different formats, depending on the temporal coverage needed (see Table 1 below). Additionally, in 2017 the specification of the product was updated to include new analyses that were available. Since March 2020, only the most recent dataset ('METOFFICE-GLO-SST-L4-NRT-OBS-GMPE-V1') is disseminated to users. GMPE was not available via CMEMS between 1 January 2023 and June 2025, following the contractual termination between CMEMS and UK institutes. Since June 2025, GMPE rejoined CMEMS dissemination as the new SST_GLO_PHY_L4_NRT_010_005 and is only available in the GDS2.0 format. Table 1 summarises the datasets and their availability.

Table 1. Dataset names with corresponding temporal coverage and GHRSST format for GMPE product

Dataset	Temporal Coverage	Format	Availability
METOFFICE-GLO-SST-L4-NRT-OBS-GMPE	17/09/2009 until 28/05/2012	GDS1.6 (old GHRSST format)	Not available online
METOFFICE-GLO-SST-L4-NRT-OBS-GMPE-V2	16/02/2012 until 26/11/2017	GDS2.0 (new GHRSST format)	Not available online
METOFFICE-GLO-SST-L4-NRT-OBS-GMPE-V3	16/09/2017 until 31/12/2022	GDS2.0	Not available online
cmems_obs-sst_glo_phy-temp_nrt_P1D-m	25/06/2025; ongoing	GDS2.0	Available for download

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	Global Ocean GMPE Sea Surface Temperature Multi Product Ensemble
Geographical coverage	Global : 180°W-180°E ; 89°S – 90°N
Variables	• analysed_sst
Product Type	Near Real Time
Available time series	From 01/01/2026 onwards
Temporal resolution	Daily
Update Frequency	Daily
Target delivery time	daily at 3:30 pm UTC
Horizontal resolution	1/4 degree
Number of vertical levels	1
Format	NetCDF4

b) Production System Description

The GHRSST Multi-Product Ensemble (GMPE) system is run daily at the Met Office (Matin, et al., 2012). It is run at 13:15 UTC and products are available to users by 15:30 UTC. It produces an ensemble field at ¼ degree for the global ocean, which is updated daily. Rolling archive of 2 years is available.

Figure 1 shows the different steps for the creation of the GMPE product at the Met Office. Each step of this processing is described below.

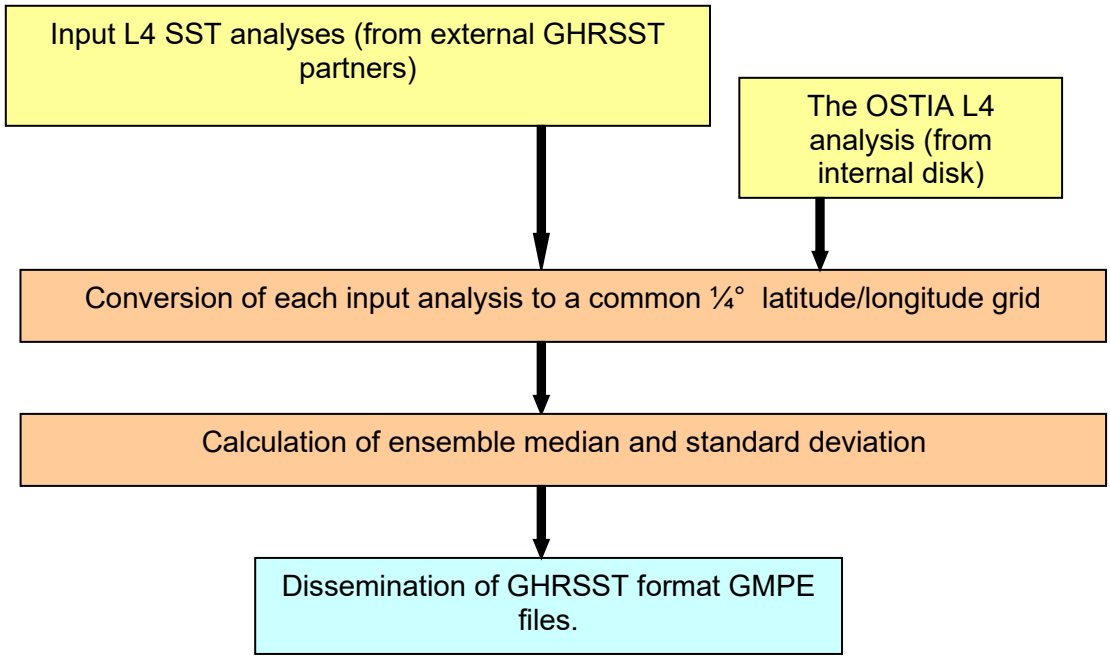

Figure 1. Schematic diagram of the GMPE processing chain at the Met Office.

Table 2 shows the basic information of the L4 analyses included in the current version of GMPE system. Please not that not all data are available at any given time, but the space is retained in the files so that each analysis is at a consistent position in the ensemble.

Table 2. List of L4 SST analyses contributing to GMPE system.

Product[1]	**Production Centre**
CMC 0.1°	Canadian Meteorological Center, Canada
DMI	Danish Meteorological Institute, Denmark
FNMOC	Fleet Numerical Meteorology and Oceanography Centre, USA
GAMSSA	Bureau of Meteorology, Australia
G1SST	NASA Jet Propulsion Laboratory, USA
MGDSST	Japan Meteorological Agency, Japan
MUR	NASA Jet Propulsion Laboratory, USA
NAVO K10	Naval Oceanographic Office, USA
ODYSSEA	Ifremer, France
OISST V2 AVHRR	National Oceanic and Atmospheric Administratio, USA

[1] Abbreviations for SST products are used as the name of the product.

OSPO Night	NOAA Office of Satellite Products and Operations, USA
OSTIA	Met Office, UK
RSS MW	Remote Sensing Systems, USA
RSS MW/IR	Remote Sensing Systems, USA

Further details of the contributing analyses are provided in the Quality Information Document for this product.

Each input L4 analysis is on a different grid. These products are re-gridded to a common ¼ ° resolution latitude/longitude grid using area averaging.

At each of the ¼ ° resolution latitude/longitude grid point, the median and standard deviation of the input SST values are calculated and output to a netCDF file.

In addition to the GMPE median field, a number of additional fields are provided for user to fully explore the GMPE product.

1. The number of analyses contributing to the ensemble at each grid point.
2. The identifier of the analysis which provides the ensemble median at each grid point.
3. The anomaly of each analysis from the median value at each grid point.
4. The horizontal gradients of each contributing analysis, and of the GMPE median. These are the averaged north-south/east-west gradients calculated using centred differencing on the original input analysis grid (i.e. before interpolation to the ¼° grid).

Figure 2 shows examples of the output fields from GMPE product on 31st May 2011. More details on the fields and evolution of the product can be found in the Quality Information Document.

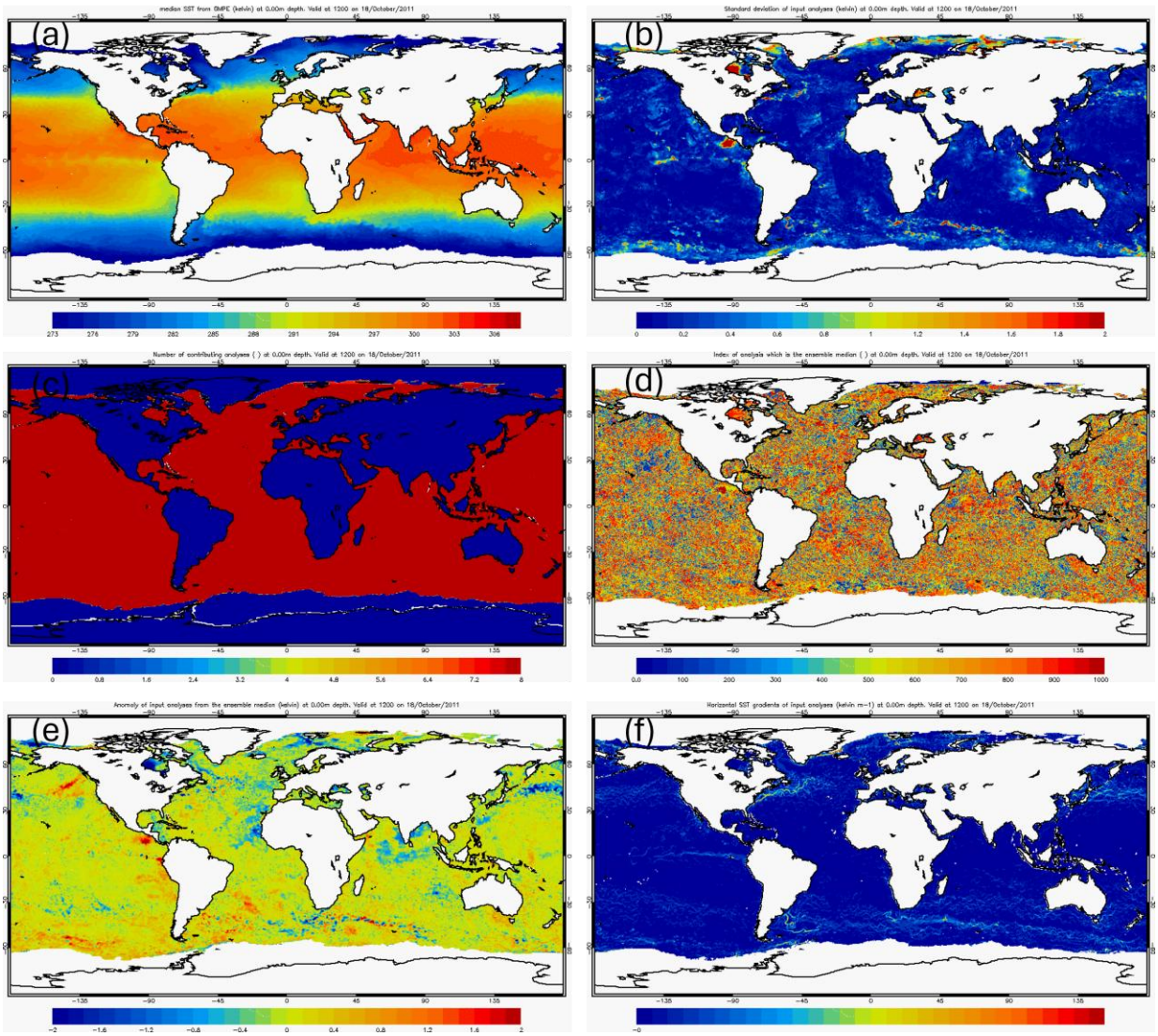

Figure 2. Examples of outputs from GMPE product on 31st May 2011 : (a) ensemble median SST, (b) ensemble standard deviation, (c) number of analyses contributing to ensemble, (d) index of analysis providing the median, (e) anomaly of one of the input analyses from the median, (f) horizontal gradient of one of the input analyses.

c) Dataset details

The GMPE products are delivered on a regular lat/lon grid, from 180 W to 180 E and 90 S to 90 N (at a ¼ ° horizontal resolution), in netCDF format. The GMPE product format specification is described in detail in the GHRSST GDS 2.0 specification. Annex 1 provides an example of the netCDF file header.

Product Lines	SST_GLO_PHY_L4_NRT_010_005
Geographical coverage	Global
Datasets: cmems_obs-sst_glo_phy-temp_nrt_P1D-m	
Product Type	Analysis
Update frequency	Dait

Available time series	From 01/01/2026 onwards
Target delivery time	Daily at 3:30 pm (15:30) UTC
Temporal resolution	Daily mean

Dataset : cmems_obs-sst_glo_phy-temp_nrt_P1D-mcontains the 3D daily mean fields: analysed_sst, standard_deviation, analysis_number, median_type, anomaly_fields, gradient_fields, field_name
analysed_sst [Kelvin] Median SST from GMPE sea_surface_foundation_temperature
standard_deviation [Kelvin] Standard deviation of input analyses
analysis_number Number of contributing analyses
median_type Index of analysis which is the ensemble median
anomaly_fields [Kelvin] Anomaly of input analyses from the ensemble median
gradient_fields [Kelvin m^{-1}] Horizontal SST gradients of input analyses

d) Additional information on some parameters

None required.

e) Processing information

Product disseminating schedule :

Type of product (forecast/ hindcast)	Forecast run (base time)	Product temporal coverage (hours)	Product disseminating scheduled time	Dissemination day
analysis	NA	24	3 :30 pm (15 :30)	Every day

f) Catalogue updating

Every day the analysis file is delivered to the catalogue and a total of 2-year rolling archive for this product is available.

g) Time averaging

Dataset frequency	Time averaging	Forecast lead time
daily	daily means over a day (midnight to midnight, centered at noon).	N/A

h) Product use and impact

SST_GLO_PHY_L4_NRT_010_005 product is based on the GMPE (GHRSST Multi Product Ensemble) system. It takes L4 SST analyses produced by various centres to create an ensemble median SST and associated statistics over the global ocean. Each analysis that contributes to GMPE uses a different subset of observational data from satellite and in-situ flatforms. Each analysis also chooses different algorithms and data assimilation schemes in the processing chain. Although all contributing analyses are "SST products", they are valid for different depths and time periods (within the day). For example, Operational Sea Surface Temperature and Sea Ice Analysis (OSTIA) is a foundation SST product (i.e. the SST free of diurnal variability) so is valid immediately before sunrise. Other products are daily means in that they use data throughout the day, but this does not necessarily ensure that a daily mean is produced. Therefore, the GMPE ensemble median SST is not strictly valid at a particular depth or time during the day.

It is used primarily as a monitoring tool for analysis data producers to assess the consistency between various SST analysis products. The standard deviation product can be used as an estimate of the confidence to be placed in each product contributing to the ensemble. It should be noted that the source of the GMPE median can vary from point to point, and therefore the GMPE median is not necessarily a consistent analysis (neither spatially nor temporally). Thus, this product is not suitable for uses such as boundary conditions to an atmosphere model or climate monitoring.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

An exhaustive description of the files nomenclature when downloaded through the web portal is available here : Files Nomenclature and Format

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

5) FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
cmems_obs-sst_glo_phy-temp_nrt_P1D-m	YYYYMMDD120000-UKMO-L4_GHRSST-SSTfnd-GMPE-GLOB-v02.0-fv03.1.nc	0.0096

6) STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

7) REFERENCES

Quality Information Document

https://documentation.marine.copernicus.eu/QUID/CMEMS-SST-QUID-010-005.pdf

Matin, M. et al., 2012. Group for High Resolution Sea Surface temperature (GHRSST) analysis fields inter-comparisons. Part 1: A GHRSST multi-product ensemble (GMPE). *Deep Sea Research Part II: Topical Studies in Oceanography,* Volume 77-80, pp. 21-30.

8) ANNEX

a) Structure and semantic of NetCDF maps files

cmems_obs-sst_glo_phy-temp_nrt_P1D-m:

```
netcdf \20251105120000-UKMO-L4_GHRSST-SSTfnd-GMPE-GLOB-v02.0-fv03.1 {
dimensions:
        lon = 1440 ;
        lat = 720 ;
        time = 1 ;
        fields = 17 ;
        fieldsp1 = 18 ;
        field_name_length = 50 ;
variables:
        int time(time) ;
                time:long_name = "reference time of sst field" ;
                time:standard_name = "time" ;
                time:axis = "T" ;
                time:calendar = "standard" ;
                time:units = "seconds since 1981-01-01 00:00:00" ;
        float lat(lat) ;
                lat:standard_name = "latitude" ;
                lat:axis = "Y" ;
                lat:units = "degrees_north" ;
                lat:valid_min = -90.f ;
                lat:valid_max = 90.f ;
                lat:reference_datum = "geographical coordinates, WGS84 projection" ;
        float lon(lon) ;
                lon:standard_name = "longitude" ;
                lon:axis = "X" ;
                lon:units = "degrees_east" ;
                lon:valid_min = -180.f ;
                lon:valid_max = 180.f ;
                lon:reference_datum = "geographical coordinates, WGS84 projection" ;
        int fields(fields) ;
                fields:long_name = "Fields number" ;
        int fieldsp1(fieldsp1) ;
                fieldsp1:long_name = "Fields number plus one" ;
        int field_name_length(field_name_length) ;
                field_name_length:long_name = "Length of string array for names of
inputs" ;
        char field_name(fields, field_name_length) ;
                field_name:long_name = "name of the contributing L4 analyses" ;
                field_name:units = "" ;
                field_name:comment = "" ;
        short analysed_sst(time, lat, lon) ;
                analysed_sst:long_name = "median SST from GMPE" ;
                analysed_sst:standard_name = "sea_surface_foundation_temperature" ;
                analysed_sst:units = "kelvin" ;
                analysed_sst:_FillValue = -32768s ;
```

```
                analysed_sst:add_offset = 273.15f ;
                analysed_sst:scale_factor = 0.01f ;
                analysed_sst:valid_min = -300s ;
                analysed_sst:valid_max = 4500s ;
                analysed_sst:source = "GAMSSA_28km, K10_SST, AVHRR_AMSRE_OI,
AVHRR_OI, OSTIA, mw_ir_OI, FNMOC, MGDSST, CMC, RTG, tmi_amsre_OI, DMI_OI,
MUR, G1SST, OSPO_Night, CMC0.1, ODYSSEA" ;
        short standard_deviation(time, lat, lon) ;
                standard_deviation:long_name = "Standard deviation of input analyses" ;
                standard_deviation:units = "kelvin" ;
                standard_deviation:_FillValue = -32768s ;
                standard_deviation:add_offset = 0.f ;
                standard_deviation:scale_factor = 0.01f ;
                standard_deviation:valid_min = -32000s ;
                standard_deviation:valid_max = 32000s ;
                standard_deviation:comment = "" ;
                standard_deviation:source = "GAMSSA_28km, K10_SST,
AVHRR_AMSRE_OI, AVHRR_OI, OSTIA, mw_ir_OI, FNMOC, MGDSST, CMC, RTG,
tmi_amsre_OI, DMI_OI, MUR, G1SST, OSPO_Night, CMC0.1, ODYSSEA" ;
        byte analysis_number(time, lat, lon) ;
                analysis_number:long_name = "Number of contributing analyses" ;
                analysis_number:units = "" ;
                analysis_number:_FillValue = -128b ;
                analysis_number:add_offset = 0.f ;
                analysis_number:scale_factor = 1.f ;
                analysis_number:valid_min = -127b ;
                analysis_number:valid_max = 127b ;
                analysis_number:comment = "" ;
                analysis_number:source = "GAMSSA_28km, K10_SST,
AVHRR_AMSRE_OI, AVHRR_OI, OSTIA, mw_ir_OI, FNMOC, MGDSST, CMC, RTG,
tmi_amsre_OI, DMI_OI, MUR, G1SST, OSPO_Night, CMC0.1, ODYSSEA" ;
        short median_type(time, lat, lon) ;
                median_type:long_name = "Index of analysis which is the ensemble
median" ;
                median_type:units = "" ;
                median_type:_FillValue = -32768s ;
                median_type:add_offset = 0.f ;
                median_type:scale_factor = 1.f ;
                median_type:valid_min = -3000s ;
                median_type:valid_max = 3000s ;
                median_type:comment = "" ;
                median_type:source = "GAMSSA_28km, K10_SST, AVHRR_AMSRE_OI,
AVHRR_OI, OSTIA, mw_ir_OI, FNMOC, MGDSST, CMC, RTG, tmi_amsre_OI, DMI_OI,
MUR, G1SST, OSPO_Night, CMC0.1, ODYSSEA" ;
        short anomaly_fields(time, fields, lat, lon) ;
                anomaly_fields:long_name = "Anomaly of input analyses from the
ensemble median" ;
                anomaly_fields:units = "kelvin" ;
                anomaly_fields:_FillValue = -32768s ;
                anomaly_fields:add_offset = 0.f ;
                anomaly_fields:scale_factor = 0.01f ;
                anomaly_fields:valid_min = -3000s ;
                anomaly_fields:valid_max = 3000s ;
                anomaly_fields:comment = "" ;
```

```
                anomaly_fields:source = "GAMSSA_28km, K10_SST, AVHRR_AMSRE_OI,
AVHRR_OI, OSTIA, mw_ir_OI, FNMOC, MGDSST, CMC, RTG, tmi_amsre_OI, DMI_OI,
MUR, G1SST, OSPO_Night, CMC0.1, ODYSSEA" ;
        short gradient_fields(time, fieldsp1, lat, lon) ;
                gradient_fields:long_name = "Horizontal SST gradients of input analyses"
;
                gradient_fields:units = "kelvin m-1" ;
                gradient_fields:_FillValue = -32768s ;
                gradient_fields:add_offset = 0.f ;
                gradient_fields:scale_factor = 1.e-05f ;
                gradient_fields:valid_min = -3000s ;
                gradient_fields:valid_max = 3000s ;
                gradient_fields:comment = "" ;
                gradient_fields:source = "GAMSSA_28km, K10_SST, AVHRR_AMSRE_OI,
AVHRR_OI, OSTIA, mw_ir_OI, FNMOC, MGDSST, CMC, RTG, tmi_amsre_OI, DMI_OI,
MUR, G1SST, OSPO_Night, CMC0.1, ODYSSEA, GMPE median" ;

// global attributes:
                :Conventions = "CF-1.4" ;
                :title = "Global SST Ensemble, L4 GMPE, 0.25deg daily (METOFFICE-GLO-
SST-L4-NRT-OBS-GMPE-V3)" ;
                :summary = "An ensemble product with input from a number of L4 SST
analyses" ;
                :references = "Martin et al., Deep Sea Research II, 2011" ;
                :institution = "UKMO" ;
                :history = "NULL" ;
                :comment = "" ;
                :license = "These data are available free of charge under the CMEMS data
policy" ;
                :id = "UKMO-L4LRens-GLOB-GMPE" ;
                :naming_authority = "org.ghrsst" ;
                :product_version = "3.6.3" ;
                :uuid = "dc0c5b25-93bf-4943-aba1-7f0de9109620" ;
                :gds_version_id = "2.0" ;
                :netcdf_version_id = "4.9.0" ;
                :date_created = "20251106T131612Z" ;
                :start_time = "20251105T000000Z" ;
                :time_coverage_start = "20251105T000000Z" ;
                :stop_time = "20251106T000000Z" ;
                :time_coverage_end = "20251106T000000Z" ;
                :file_quality_level = 3 ;
                :source = "GAMSSA_28km, K10_SST, AVHRR_AMSRE_OI, AVHRR_OI,
OSTIA, mw_ir_OI, FNMOC, MGDSST, CMC, RTG, tmi_amsre_OI, DMI_OI, MUR, G1SST,
OSPO_Night, CMC0.1, ODYSSEA" ;
                :platform = "" ;
                :sensor = "" ;
                :metadata_conventions = "Unidata Observation Dataset v1.0" ;
                :metadata_link = "not available" ;
                :keywords = "Oceans > Ocean Temperature > Sea Surface Temperature" ;
                :keywords_vocabulary = "NASA Global Change Master Directory (GCMD)
Science Keywords" ;
                :standard_name_vocabulary = "NetCDF Climate and Forecast (CF)
Metadata Convention" ;
                :westernmost_longitude = -180.f ;
```

```
                :easternmost_longitude = 180.f ;
                :southernmost_latitude = -90.f ;
                :northernmost_latitude = 90.f ;
                :spatial_resolution = "0.25 degree" ;
                :geospatial_lat_units = "degrees north" ;
                :geospatial_lat_resolution = "0.25 degree" ;
                :geospatial_lon_units = "degrees east" ;
                :geospatial_lon_resolution = "0.25 degree" ;
                :acknowledgment = "Please acknowledge the use of these data with the
following statement: These data were provided by GHRSST, Met Office and CMEMS" ;
                :creator_name = "Met Office as part of CMEMS" ;
                :creator_email = "servicedesk.cmems@mercator-ocean.eu" ;
                :creator_url = "http://marine.copernicus.eu" ;
                :project = "Group for High Resolution Sea Surface Temperature" ;
                :publisher_name = "GHRSST Project Office" ;
                :publisher_url = "http://www.ghrsst.org" ;
                :publisher_email = "ghrsst-po@nceo.ac.uk" ;
                :processing_level = "GMPE" ;
                :cdm_data_type = "grid" ;
}
```